

STRUCTURE SIZE IN C

Question

Create a structure variable student and estimate the size of the structure manually. Then write a C-program to find the size of a structure using sizeof operator and observe both the output. Are they same or different? Justify your answer.

C Program

```
#include <stdio.h>

struct student
{
    int rollno;
    char grade;
    float marks;
};

int main()
{
    struct student s;
    printf("Size of structure = %zu bytes\n", sizeof(s));
    return 0;
}
```

Manual Calculation

int = 4 bytes

char = 1 byte

float = 4 bytes

Total = 4 + 1 + 4 = 9 bytes

Program Output

Size of structure = 12 bytes

Conclusion / Justification

No, they are different. The difference occurs because of structure padding and alignment. The compiler inserts extra unused bytes between members or at the end of the structure so that the data is properly aligned in memory for faster access by the CPU.

Therefore, the manually calculated size is 9 bytes, whereas the size reported by sizeof is 12 bytes.